

E-Commerce Sales Analysis

Business Story & Key Insights

1. Business Context

The goal of this analysis was to understand product performance, sales dynamics over time, and customer behavior using transactional e-commerce data.

The project focuses on identifying revenue drivers, analyzing purchasing patterns, and evaluating customer activity through SQL analysis and Power BI visualization.

2. Product Performance Analysis

Revenue Concentration Across Products

Product analysis shows that revenue is not evenly distributed across all items.

A small number of high-performing SKUs generate a disproportionately large share of total revenue. While unit sales remain relatively stable across many products, revenue differences are mainly influenced by product value and transaction size.

Key Insight

High-value products have a significant impact on overall business performance, making product prioritization and inventory decisions important for revenue optimization.

3. Product Structure Analysis

Revenue vs Sales Volume Relationship

The analysis of units sold and average transaction value shows that most products operate within a similar sales volume range.

However, only a limited number of products combine high sales volume with high revenue contribution, making them important revenue drivers.

Key Insight

Revenue performance is not determined only by sales volume. Product value and pricing structure play a major role in business contribution.

4. Time-Based Sales Analysis

Revenue Trends and Seasonality

The time analysis shows relatively stable revenue patterns over the analyzed period.

No strong long-term growth or decline trend was identified. However, recurring revenue decreases in February indicate possible seasonal demand changes or post-holiday normalization effects.

Key Insight

The business demonstrates stable demand with moderate seasonal fluctuations rather than strong volatility.

5. Customer Behavior Analysis

Purchase Frequency and Customer Value

Customer analysis reveals a positive relationship between purchase frequency and generated revenue.

Customers with higher purchasing frequency contribute significantly more revenue compared to occasional buyers.

Key Insight

Repeat purchases are a major driver of customer lifetime value, highlighting the importance of retention strategies.

6. Customer Activity Analysis (Recency)

Customer Engagement and Retention

Recency analysis indicates that most customers have made purchases recently, with a large proportion of customers remaining active.

Only a small percentage of customers fall into long-term inactive groups.

Key Insight

The customer base demonstrates strong activity levels and relatively low inactivity risk.

7. Final Business Conclusion

The analysis shows that business performance is mainly driven by:

- repeat customer activity;
- a small number of high-value products;
- stable purchasing patterns.

Revenue is not primarily dependent on strong seasonality or one-time transactions. Instead, consistent customer engagement and key revenue-generating products create business stability.

Interview Summary

If asked:

"What did you learn from this project?"

Main findings:

- Product concentration: a small number of SKUs drive a large revenue share.
- Stable demand: sales show limited volatility over time.
- Customer loyalty: repeat customers contribute significantly to revenue.
- Weak seasonality: no strong seasonal dependency was identified.
- High-value products: certain products have a disproportionate business impact.